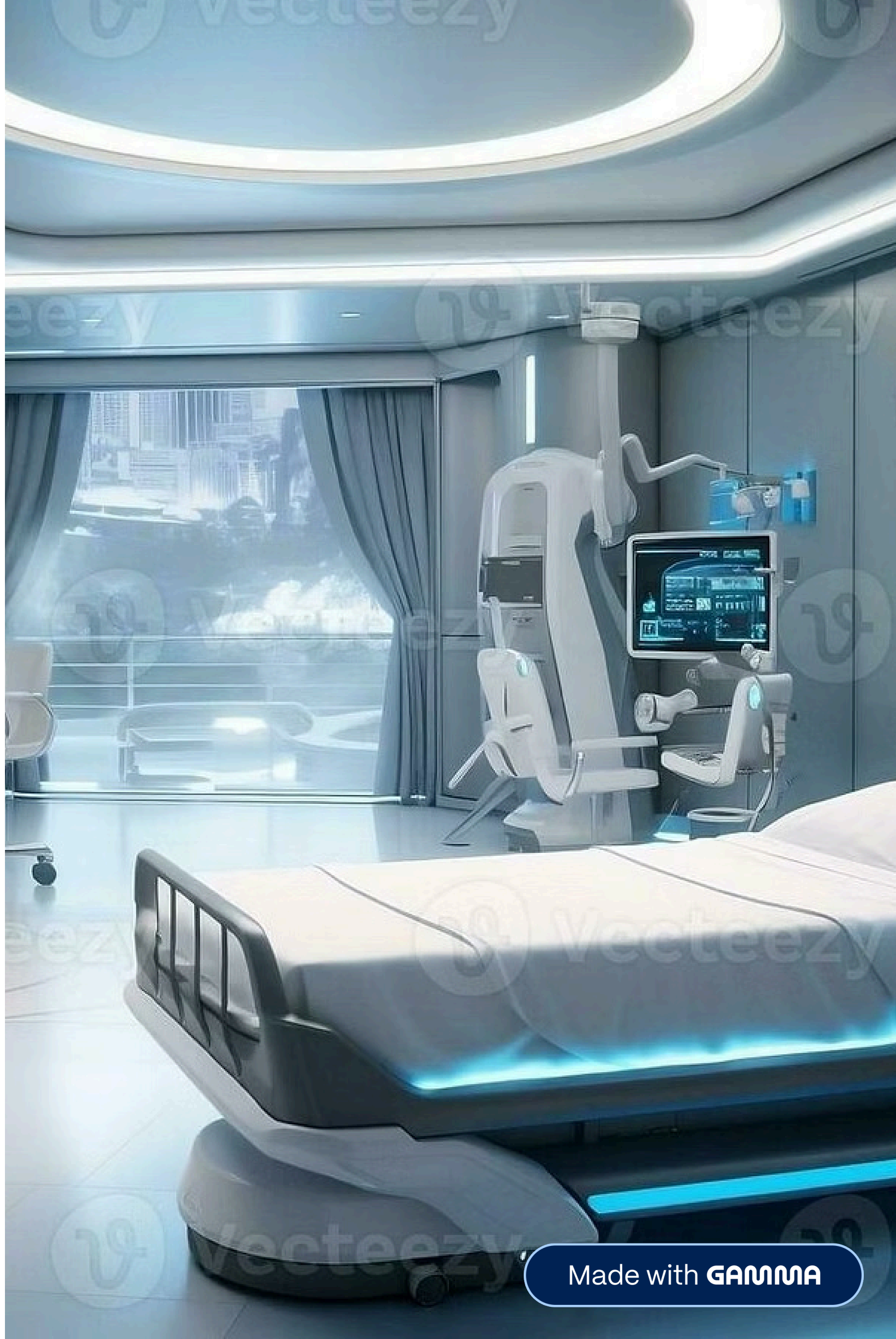
DischargeAssist AI

AI-Powered Clinical Summary and Patient Discharge Assistance System

DischargeAssist AI is an intelligent healthcare summarization system that transforms raw clinical notes into structured discharge summaries, medication guidance, follow-up instructions, and patient-friendly explanations with readability analysis, confidence scoring, and multilingual support.



Problem We Are Solving

Unstructured Clinical Notes

Discharge information is written in lengthy, unstructured formats that are difficult to navigate and extract critical information from efficiently.

Time-Consuming Manual Work

Clinicians spend excessive time manually summarizing and organizing patient discharge documentation, reducing time available for direct patient care.

Complex Medical Jargon

Patients struggle to understand medical terminology and treatment instructions, leading to medication errors and missed follow-up appointments.

Poor Communication Quality

Critical follow-up details and medication instructions get buried in text, compromising patient safety and care continuity after discharge.

Objective of DischargeAssist AI

1. **Generate SOAP Notes**

   Automatically structure clinical information into standardized SOAP format (Subjective, Objective, Assessment, Plan)

2. **Extract Medications**

   Identify and organize all prescribed medications with dosages, frequencies, and instructions

3. **Draft Follow-Up Instructions**

   Create clear guidance for post-discharge care, appointments, and warning signs

4. **Create Patient-Friendly Summaries**

   Simplify medical content for patient comprehension at appropriate reading levels

5. **Improve Readability**

   Adapt content to target 6th-grade reading level for universal patient understanding

6. **Verify Confidence & Factual Support**

   Ensure generated content matches source notes and flags potential hallucinations

7. **Support Multilingual Output**

   Translate summaries into regional languages for diverse patient populations

End-to-End Workflow

The system processes clinical notes through a modular AI pipeline with validation checkpoints:

Raw Clinical Note Input

Unstructured clinical documentation entered by healthcare provider

Text Preprocessing

Normalization and cleaning of input text

SOAP Generation

Structured SOAP note creation

Medication Extraction

Drug identification and classification

Follow-Up Drafting

Post-discharge care planning

Patient Summary Generation

Consumer-friendly explanation creation

Readability Simplification

Adapt to target reading level

Confidence Scoring

Factual accuracy verification

Translation

Multilingual output generation

Database Storage

History and audit logging

Architecture of the Project

Frontend Interface

dischargeassist.html – User interface for note input and summary display

Backend API

main.py using FastAPI – Request handling and pipeline orchestration

LLM Engine

llm_engine.py with Gemini API – Core AI generation and reasoning

Processing Modules

SOAP, medications, follow-up, readability, confidence, translation

Data Storage

SQLite via database.py – Audit logs and summary persistence

This modular architecture enables independent development, testing, and evaluation of each system component while maintaining clear data flow and integration points.

Key Files Used in the Project

Pipeline & API

main.py – Full pipeline orchestration and API endpoints

models.py – Input/output data models

Text Processing

preprocessor.py – Cleans and normalizes note text

prompt_templates.py – All LLM prompts

Document Generation

soap_formatter.py – SOAP note generation

medication_explainer.py – Medication extraction

followup_drafter.py – Follow-up generation

Patient Communication

patient_simplifier.py – Patient-friendly summary

readability.py – Grade-level simplification

multilingual.py – Translation

Validation

confidence_scorer.py – Confidence and hallucination checks

Persistence

database.py – Audit logs and summary storage

Main Logic and AI Features

The intelligence combines LLM generation with validation layers for safer, more useful outputs:

- Structured SOAP format conversion
- Medication extraction and classification
- Follow-up guidance generation
- Patient summary simplification
- Flesch-Kincaid readability measurement
- Source content matching verification
- Regional language translation

Key Differentiator: Validation layers make this system more useful and safer than simple text generation by adding confidence checks and readability adaptation.

Parameters and Technical Choices

6

Target Readability Grade

Optimized for universal patient comprehension

3

Maximum Refinement Passes

Balances quality with processing time

0

LLM Temperature

Zero temperature for factual, non-creative outputs

2048

Maximum Output Tokens

Adequate context window for detailed summaries

45s

Request Timeout

Prevents hanging requests during processing

4

Retry Attempts

Handles transient API failures gracefully

These settings prioritize stability, factual accuracy, patient comprehension, and healthcare suitability over creative outputs.

Evaluation of the Project

✓ Benefits

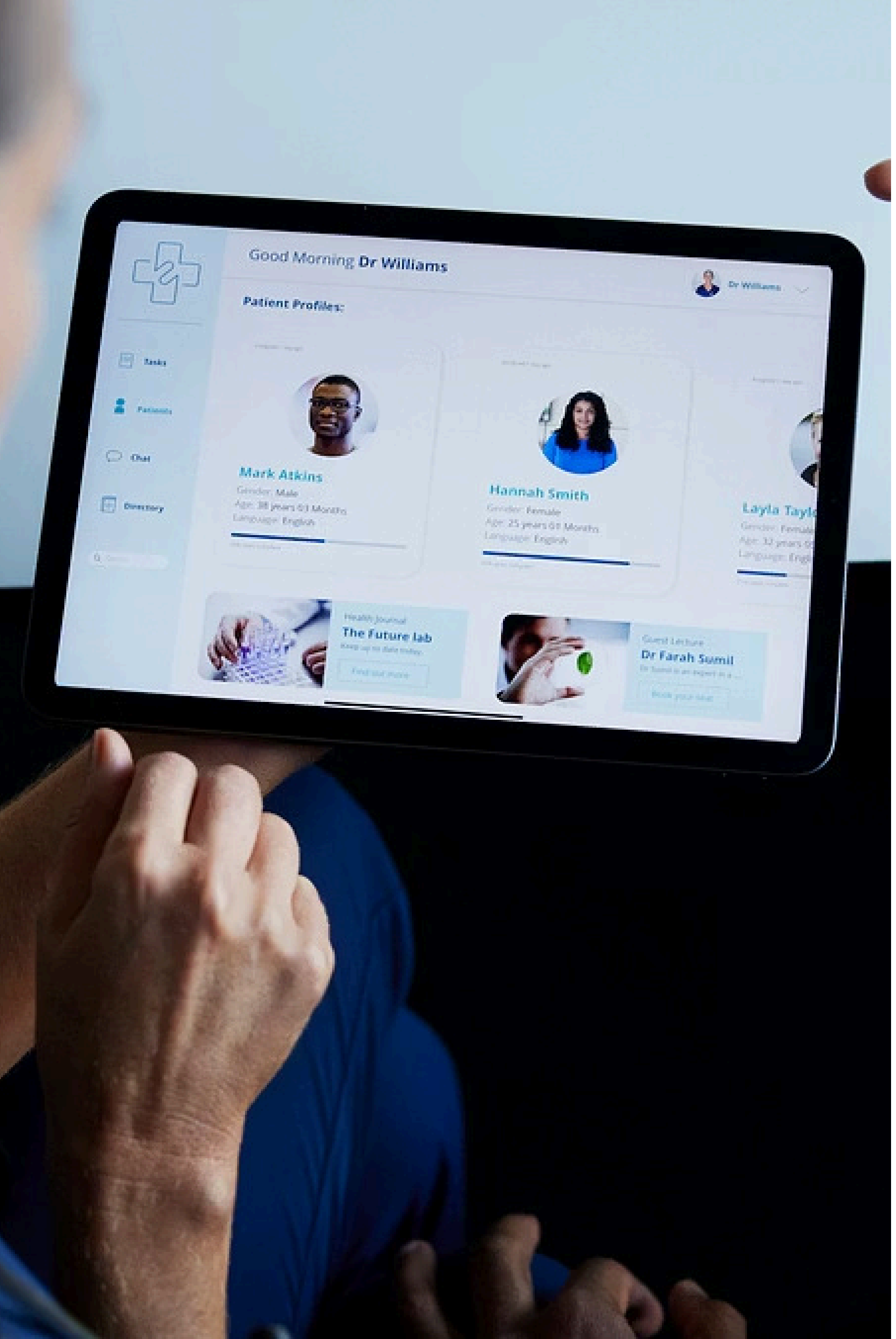
- Reduces clinician documentation burden
- Improves discharge communication clarity
- Enhances patient understanding
- Adds confidence and readability validation
- Supports multilingual patient populations

⚠️ Limitations

- Dependent on underlying LLM quality
- Not a replacement for clinical review
- Translation accuracy varies by language
- Confidence scoring has false positives
- Requires clean input documentation

→ Future Scope

- EHR and PDF upload integration
- Enhanced hallucination detection
- Additional language support
- Hospital system integration
- Clinical validation studies



Conclusion

> DischargeAssist AI demonstrates how AI can enhance healthcare communication while maintaining appropriate human oversight for patient safety.

This modular healthcare AI system transforms raw clinical notes into structured, readable, and patient-centered discharge outputs. By combining clinical summarization, medication extraction, follow-up generation, readability adaptation, confidence scoring, and multilingual support in one pipeline, the system addresses critical gaps in discharge communication.

Thank You